

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 18: Wastewater Story

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	18 — Wastewater Story
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps target the most common Wastewater-Story confusions: treating 'cleaner' as 'drinkable'; swapping sludge (sinks) with scum (floats); mixing up aerobic aeration with anaerobic digestion; assuming bar screens or settling remove dissolved/microscopic impurities; thinking grit removal and sludge settling are the same step; believing pouring oil/chemicals/medicines down the drain is harmless; and confusing which by-product (biogas vs manure) comes from which process. Do not write on this paper — mark answers on your answer sheet.

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1. A chemistry teacher passes a sample of raw sewage through filter paper and finds the clear filtrate still contains dissolved nitrates. This shows that nitrates in sewage are best classified as which kind of impurity?
 - (A) Suspended organic matter that the filter paper traps
 - (B) Living microbes that reproduce on the filter paper
 - (C) Floating scum that stays on top of the filtrate
 - (D) Dissolved inorganic impurities that pass through a simple filter
 2. Two jars of sewage are left to stand. Jar X has its top oily layer scraped off; Jar Y has the muck at its bottom scooped out. What was removed from each jar?
 - (A) Sludge from Jar X and scum from Jar Y
 - (B) Grit from Jar X and biogas from Jar Y
 - (C) Scum from Jar X and sludge from Jar Y
 - (D) Manure from Jar X and scum from Jar Y
 3. At a treatment plant the operator deliberately makes the wastewater flow very slowly through the grit chamber. What is the direct purpose of slowing the flow here?

- (A) To give heavy sand, grit and pebbles time to sink to the bottom
- (B) To let dissolved nitrates settle out of the water
- (C) To let aerobic bacteria multiply in the chamber
- (D) To allow oil and grease time to rise and be skimmed

4. If the bar screens at a plant were completely removed, which problem would appear FIRST as sewage moves downstream?

- (A) Dissolved nitrates would suddenly double in the water
- (B) Rags, plastics and sticks would clog and damage the pumps and pipes ahead
- (C) Aerobic bacteria in the aeration tank would die immediately
- (D) The released water would instantly become safe to drink

5. A student writes: 'Aerobic bacteria in the aeration tank need air, while the bacteria in the sludge digester work without air.' Which statement correctly judges this?

- (A) Correct — aeration uses oxygen-needing bacteria; the digester uses anaerobic bacteria
- (B) Wrong — both tanks use bacteria that need plenty of air
- (C) Wrong — the aeration tank works without air and the digester needs air
- (D) Wrong — neither process uses bacteria at all

6. Which by-product is correctly paired with the process that makes it?

- (A) Biogas — produced when air is bubbled through the aeration tank
- (B) Manure — skimmed off the settling tank as floating scum
- (C) Biogas — given off by the bar screens removing debris
- (D) Biogas — produced by anaerobic digestion of sludge

7. A worker claims the clarified water leaving a treatment plant is now pure enough to drink straight from the outlet. The best response is:

- (A) Right — treatment removes every last impurity and germ
- (B) Wrong — the water is much cleaner but is usually still not fit for drinking
- (C) Right — once it leaves the plant it equals bottled drinking water
- (D) Wrong — the outlet water is actually dirtier than raw sewage

8. Place these four removal events in the order a drop of sewage actually experiences them: (i) floating oil skimmed as scum, (ii) plastic bags caught, (iii) heavy sand settled out, (iv) remaining organic matter eaten by aerobic bacteria.

- | | |
|-----------------------|-----------------------|
| (A) i → ii → iii → iv | (B) ii → iii → i → iv |
| (C) iv → iii → ii → i | (D) iii → ii → i → iv |

9. Why are sand and grit deliberately taken out in their own chamber BEFORE the wastewater reaches the settling tank?

- (A) Because sand is the most poisonous contaminant in sewage
- (B) Because sand is the raw material needed to make biogas

(C) So the hard, abrasive particles do not wear out equipment or get mixed into the sludge handled later

(D) Because the settling tank is unable to hold any solids at all

10. A family pours a litre of used cooking oil down the kitchen sink every week. Which chain of effects best matches the chapter's warning?

(A) The oil is the favourite food of aerobic bacteria and speeds up cleaning

(B) The oil makes the treated water taste sweet and safe to drink

(C) The oil instantly turns into useful manure inside the sewer

(D) The oil coats and clogs the pipes and makes treatment harder

11. In the settling tank the scraper at the bottom and the skimmer at the top each do a different job. What does the bottom scraper handle?

(A) It lifts the floating oily scum off the surface

(B) It bubbles air through the water to feed bacteria

(C) It pushes the settled sludge toward removal at the bottom

(D) It screens out large plastic bags and rags

12. The aeration tank's air pump fails and stays off for two days. What is the most likely consequence?

(A) Aerobic bacteria struggle for oxygen, so leftover organic waste is poorly broken down

(B) Aerobic bacteria thrive even more without the disturbance of air

(C) Biogas is produced in the tank instead of clean water

(D) The dissolved organic matter is immediately filtered out by gravity

13. Trace the correct path of dirty water: from a house it goes to the _____, then to the _____, and finally the cleaned water goes to a _____.

(A) sewer; treatment plant; river or water body

(B) treatment plant; sewer; river or water body

(C) river; sewer; treatment plant

(D) biogas plant; bar screen; home tap

14. Which contaminant in sewage is the one that is actually ALIVE and can directly make people ill?

(A) Dissolved phosphate salts

(B) Suspended particles of sand

(C) Disease-causing micro-organisms

(D) Floating drops of cooking oil

15. A town with no underground sewers wants a low-cost way for a single household to handle toilet waste on the spot. Which option fits?

(A) A septic tank built near the house

(B) A full-scale municipal sewage treatment plant for that one house

(C) A bar screen installed under the toilet

(D) A grit-and-sand chamber for each home

16. Activated sludge from the aeration tank settles at the bottom of the tank. What does the chapter say is done with most of this settled bacterial sludge?

- (A) A large part is returned to the aeration tank to keep the bacterial action going
- (B) It is poured directly into the river untreated
- (C) It is skimmed off the top as scum
- (D) It is screened out by the bar screens

17. A factory quietly pipes toxic chemicals straight into the public sewer. Why is this especially harmful to the treatment process itself?

- (A) The chemicals make the bar screens catch debris faster
- (B) The chemicals can kill the helpful bacteria the plant relies on to clean the water
- (C) The chemicals turn the sewage instantly into safe manure
- (D) The chemicals increase the town's drinking-water supply

18. Which statement about what the bar screens can and cannot do is correct?

- (A) They catch large objects like cans and rags but leave dissolved and microscopic impurities behind
- (B) They remove dissolved nitrates along with the large objects
- (C) They kill all the disease-causing microbes in the sewage
- (D) They make the sewage completely free of every impurity

19. Why does good community sanitation, such as proper toilets and covered sewers, reduce diseases like cholera and typhoid?

- (A) It adds extra nutrients that strengthen the drinking water
- (B) It keeps disease-carrying sewage from mixing with the water people drink
- (C) It increases the number of microbes in household taps
- (D) It makes biogas burn with a brighter flame

20. Rank what is removed earliest to latest in the plant: (i) big floating objects, (ii) remaining dissolved/organic matter by bacteria, (iii) heavy grit, (iv) settled sludge and floating scum.

- (A) ii → iv → iii → i
- (B) i → iii → iv → ii
- (C) iii → i → iv → ii
- (D) iv → i → iii → ii

21. After the closed digester finishes acting on the sludge, the leftover digested sludge is dried and used as _____, while the gas released is used as _____.

- (A) scum; a disinfectant for drinking water
- (B) fuel; manure
- (C) manure; fuel
- (D) grit; a building material

22. Which of the following is NOT an organic contaminant normally carried by household sewage?

- (A) Leftover food scraps washed off plates
- (B) Human faeces from toilets
- (C) Dissolved iron salts from a metal factory
- (D) Used cooking oil from the kitchen

- 23.** A student says, 'Once the water passes the grit chamber, all its solids are gone.' What is the correct judgement?
- (A) Wrong — only heavy sand and grit are removed; finer solids, scum and dissolved matter still remain
 - (B) Right — the grit chamber removes every solid completely
 - (C) Right — after grit removal only pure water is left
 - (D) Wrong — the grit chamber actually adds new solids to the water
- 24.** Why does the chapter advise against flushing leftover medicines and strong chemicals down the toilet?
- (A) They give the aerobic bacteria extra energy to work faster
 - (B) They turn the sewage directly into clean drinking water
 - (C) They help the bar screens trap large objects more efficiently
 - (D) They can harm the bacteria that the treatment plant depends on to clean the water
- 25.** Both the sludge digester and the aeration tank use bacteria, yet they are kept very different. Which contrast is correct?
- (A) The digester is supplied air while the aeration tank is sealed off from air
 - (B) Both are sealed off from air and use the same anaerobic bacteria
 - (C) Both are aerated and rely entirely on aerobic bacteria
 - (D) The digester is sealed and air-free for anaerobic bacteria; the aeration tank is supplied air for aerobic bacteria
- 26.** Open defecation in fields near a village is dangerous mainly because:
- (A) The waste quickly becomes safe biogas in the open air
 - (B) It increases the village's supply of clean drinking water
 - (C) It feeds the aerobic bacteria in the village taps
 - (D) Rain can wash the waste and its germs into water sources, spreading disease
- 27.** An engineer wants to remove fine suspended solids that the grit chamber missed. Which tank is designed for this?
- (A) The bar screen, which catches the fine suspended solids
 - (B) The settling (sedimentation) tank, where solids settle as sludge
 - (C) The sludge digester, which sieves out suspended solids
 - (D) The aeration tank, whose air pushes solids to the bottom
- 28.** Which sequence correctly names the air condition of each biological step in the right order?
- (A) Aeration tank — without air (anaerobic); sludge digester — with air (aerobic)
 - (B) Both steps — with air (aerobic)
 - (C) Aeration tank — with air (aerobic); sludge digester — without air (anaerobic)
 - (D) Both steps — without air (anaerobic)

- 29.** A vermi-processing toilet is considered environment-friendly because it:
- (A) Flushes the waste straight into the nearest river
 - (B) Burns the waste to make biogas without any bacteria
 - (C) Pumps air through the waste to grow aerobic bacteria
 - (D) Uses earthworms to turn human waste into useful manure
- 30.** Why is the clarified water sometimes disinfected with chlorine before it is released?
- (A) To kill harmful microbes that earlier physical and biological steps did not remove
 - (B) To add nitrates that make the water richer
 - (C) To settle out the remaining heavy sand and grit
 - (D) To produce biogas from the leftover organic matter
- 31.** Sewage from a hospital, a home and a farm all enter the same sewer. What does this tell us about the sources of sewage?
- (A) Sewage comes only from household toilets and nowhere else
 - (B) Sewage is just rainwater collected from rooftops
 - (C) Sewage comes from domestic, industrial/institutional and agricultural sources together
 - (D) Sewage is produced inside the treatment plant itself
- 32.** Which observation would tell a worker that a tank is the settling tank rather than the aeration tank?
- (A) Air is being bubbled vigorously through the water to grow bacteria
 - (B) A sludge layer is forming at the bottom and an oily scum floats on top, with no air being bubbled
 - (C) Large plastic bags are being caught by metal bars
 - (D) Flammable biogas is being collected from a sealed dome
- 33.** Phosphates and nitrates dissolved in sewage are sometimes called 'nutrients'. If sewage rich in these is released untreated into a lake, the most likely effect is:
- (A) The nutrients make the lake water safe to drink
 - (B) Excess nutrients can over-feed water plants and disturb the lake's balance
 - (C) The nutrients settle out instantly as harmless sand
 - (D) The nutrients turn directly into biogas in the lake
- 34.** Which statement about scum is correct?
- (A) Scum is the heavy solid material scraped from the bottom
 - (B) Scum is the flammable gas collected from the digester
 - (C) Scum is light, floating matter such as oil and grease that is skimmed off the surface
 - (D) Scum is the dried sludge spread on fields as manure
- 35.** Why must sewage be treated rather than just dumped directly into a river?

- (A) To make the river deeper for boats
- (B) To supply construction sand to builders
- (C) To raise the temperature of the river water
- (D) To prevent water pollution and the spread of waterborne diseases

36. A plant's settling tank scraper jams and stops working. What is the most direct consequence?

- (A) Floating scum is no longer skimmed from the surface
- (B) Air stops bubbling through the aeration tank
- (C) Bar screens stop catching large debris at the entrance
- (D) Settled sludge builds up at the bottom and is not carried away for treatment

37. Which household item, if poured down the drain, is LEAST likely to harm the treatment bacteria or clog the pipes?

- (A) Leftover paint thinner
- (B) Hot used cooking oil
- (C) Plain water used to rinse a glass
- (D) Old expired medicines

38. Which correctly states what happens biologically to the two streams inside the plant?

- (A) Both the sludge and the water are treated together in one air-free tank
- (B) The sludge is digested without air, while the separate clarified water is aerated with air
- (C) The clarified water is digested without air and the sludge is aerated
- (D) Bacteria are used only on the bar screens, nowhere else

39. In a chemical toilet used at a remote campsite, why are chemicals added to the waste?

- (A) To turn the waste straight into drinking water on the spot
- (B) To grow aerobic bacteria that need a constant air pump
- (C) To break down the waste and reduce odour and germs where no sewer exists
- (D) To collect flammable biogas for cooking at the camp

40. Two samples are taken: one from the inlet of a plant (raw sewage) and one from the outlet (treated water). Which comparison is correct?

- (A) The outlet sample is dirtier than the inlet sample
- (B) Both samples are identical because treatment changes nothing
- (C) The outlet sample is guaranteed pure enough to drink
- (D) The outlet sample has far less organic matter and fewer solids than the inlet sample

41. Why does the chapter say a treatment plant alone is not enough, and good habits at home also matter?

- (A) Because dumping oils, chemicals and solids down drains can clog pipes and damage the plant's bacteria
- (B) Because the plant works better if homes pour more oil into drains
- (C) Because home habits change the order of the treatment steps
- (D) Because households are expected to disinfect water themselves

- 42.** Which step removes impurities by a purely physical/mechanical action rather than by bacteria?
- (A) The aeration tank consuming organic matter
 - (B) The sludge digester breaking down sludge
 - (C) The activated-sludge bacteria settling out
 - (D) The bar screen catching large objects
- 43.** A claim reads: 'Biogas from the digester is the same thing as the scum from the settling tank.' This is:
- (A) Right — both are the floating oily layer on the water
 - (B) Wrong — biogas is a flammable gas from anaerobic digestion, while scum is a floating oily solid skimmed earlier
 - (C) Right — both are flammable gases collected from sealed tanks
 - (D) Wrong — biogas is the bottom sludge and scum is a gas
- 44.** Why is the sludge sent to a sealed digester instead of being left in an open tank exposed to air?
- (A) So aerobic bacteria can grow faster in the open air
 - (B) So the sun can dry it instantly into manure
 - (C) So anaerobic bacteria can decompose it and produce useful biogas in the absence of air
 - (D) So large plastics in it can be screened out
- 45.** During heavy rain, lots of grit and sand enter a town's sewers. Which plant stage is most important for handling this load?
- (A) The grit-and-sand removal chamber
 - (B) The sludge digester
 - (C) The chlorination/disinfection step
 - (D) The aeration tank
- 46.** Which best explains the difference between sludge and grit, both of which settle in the plant?
- (A) Grit and sludge are the same material removed in one tank
 - (B) Grit is the floating oily layer and sludge is the gas
 - (C) Grit is heavy mineral sand removed early; sludge is mostly organic solids that settle later in the settling tank
 - (D) Grit is organic waste and sludge is clean drinking water
- 47.** A municipal worker regularly cleans street drains and public toilets. How does this help public health?
- (A) It stops drains from overflowing and keeps disease-spreading waste away from people
 - (B) It increases the number of microbes reaching homes
 - (C) It pours extra chemicals into household sinks
 - (D) It speeds up biogas burning in kitchens

- 48.** A student argues, 'Aerobic bacteria do the work in the digester and anaerobic bacteria do the work in the aeration tank.' What is wrong with this?
- (A) Nothing is wrong; the statement is exactly correct
 - (B) Only the digester part is wrong; the aeration part is correct
 - (C) Both tanks actually use the same aerobic bacteria
 - (D) It is reversed — anaerobic bacteria work in the air-free digester and aerobic bacteria work in the aerated tank
- 49.** Why is treated effluent often used to recharge groundwater rather than being wasted?
- (A) Because the effluent is pure drinking water needing no further care
 - (B) Because the effluent is dirtier than raw sewage and must be hidden
 - (C) Because recharging turns the effluent into biogas underground
 - (D) Because the cleaned water, though not for drinking, can safely return to the ground and replenish supplies
- 50.** If you observed flammable gas being safely collected from a closed dome at a plant, which stage are you watching?
- (A) The aeration tank producing oxygen for bacteria
 - (B) The bar screen releasing gas from debris
 - (C) The sludge digester producing biogas
 - (D) The settling tank giving off gas from scum
- 51.** Which is the correct early-to-late order of the materials that LEAVE the main water stream?
- (A) Large debris first, then grit, then settled sludge and floating scum, then organic matter eaten by bacteria
 - (B) Organic matter eaten by bacteria first, then sludge and scum, then grit, then large debris
 - (C) Grit first, then large debris, then organic matter, then sludge and scum
 - (D) Floating scum first, then grit, then large debris, then organic matter
- 52.** Why can pouring solid food waste and tea leaves down the sink cause trouble even though they are 'natural'?
- (A) They instantly dissolve and improve the water quality
 - (B) They can build up and choke the pipes, blocking the flow of sewage
 - (C) They turn straight into clean drinking water in the pipe
 - (D) They are eaten by aerobic bacteria living inside home taps
- 53.** A plant releases its treated water into a river. A downstream village still must not assume the water is drinkable because:
- (A) The plant deliberately makes the water dirtier before release
 - (B) Treatment makes water much cleaner but does not guarantee it is free of all germs and safe to drink

- (C) Treated water always contains more sewage than the inlet
- (D) Rivers automatically purify any water that enters them

54. Which combination of products from the whole treatment process is genuinely useful?

- (A) Plastic bags from the bar screen and grit from the chamber
- (B) Scum as a cooking fuel and grit as a fertiliser
- (C) Biogas as fuel and dried digested sludge as manure
- (D) Raw untreated sludge poured into the river as fish food

55. Why is it important that grit removal comes BEFORE the aeration tank in the sequence?

- (A) So abrasive sand does not reach and damage the aeration equipment or smother the bacteria
- (B) Because sand is needed to feed the aerobic bacteria
- (C) Because the aeration tank removes grit better than the grit chamber
- (D) Because grit must be aerated before it can settle

56. Which describes correctly what 'activated sludge' is?

- (A) The large plastic debris caught on the bar screens
- (B) The flammable gas given off by the sealed digester
- (C) The heavy sand and grit settled in the grit chamber
- (D) The settled mass of aerobic bacteria from the aeration tank that have cleaned the water

57. A town with poor sanitation reports repeated outbreaks of typhoid and cholera. The most likely root cause is:

- (A) Too much biogas is being produced in homes
- (B) Sewage is mixing with the water people drink
- (C) The treated water is being recharged into the ground
- (D) The bar screens are catching too many plastics

58. Choose the statement that correctly distinguishes where dissolved impurities are dealt with versus where large solids are dealt with.

- (A) Large solids are screened or settled mechanically, while dissolved organic matter is consumed by bacteria in aeration
- (B) Both large solids and dissolved organic matter are removed by the bar screens
- (C) Dissolved impurities are caught by the grit chamber and solids by the digester
- (D) Dissolved organic matter is skimmed off as scum and solids are aerated

59. A septic tank serves a single rural home. After solids settle inside it over months, what eventually has to be done with the accumulated sludge?

- (A) It turns into clean drinking water that the family can use
- (B) It is automatically converted into fuel piped to the kitchen

(C) It never accumulates, so the tank needs no attention at all

(D) It must be periodically pumped out and disposed of or treated, not left forever

60. Why does the chapter encourage practices like not throwing solid waste into drains and supporting proper sewage systems, rather than relying only on doctors to treat illness?

(A) Because doctors cannot treat any waterborne disease at all

(B) Because preventing sewage from contaminating water stops diseases before they start, which is better than only curing them

(C) Because clean drains produce more biogas for hospitals

(D) Because blocked drains make drinking water safer to use

Solutions — Science (Class 7), Chapter 18:

Wastewater Story — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	C	31	C	41	A	51	A
2	C	12	A	22	C	32	B	42	D	52	B
3	A	13	A	23	A	33	B	43	B	53	B
4	B	14	C	24	D	34	C	44	C	54	C
5	A	15	A	25	D	35	D	45	A	55	A
6	D	16	A	26	D	36	D	46	C	56	D
7	B	17	B	27	B	37	C	47	A	57	B
8	B	18	A	28	C	38	B	48	D	58	A
9	C	19	B	29	D	39	C	49	D	59	D
10	D	20	B	30	A	40	D	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Because the nitrates pass straight through filter paper and remain in the clear liquid, they must be dissolved, not suspended. They are mineral salts, so they are inorganic. The option calling them suspended organic matter is wrong since suspended matter would be trapped by the paper. They are chemicals, not living things, so the microbe option fails. Scum is floating oil and grease, not a dissolved salt, so that option is also wrong.

2. (C) Oil and grease are light and float, forming the layer skimmed off the top called scum, so the top layer of Jar X is scum. The heavier solids sink and are removed from the bottom as sludge, so Jar Y gives sludge. Reversing the two contradicts the fact that oil floats while solids sink. Grit is removed earlier in a separate chamber, and biogas is a gas made later, so that pairing is wrong. Manure is dried, digested sludge, not the floating oily skim, so that option is also incorrect.

- 3. (A)** Slow flow reduces turbulence so dense particles like sand and grit have time to settle by gravity. Dissolved nitrates stay dissolved no matter how slow the flow, so they cannot settle here. Aerobic bacteria are cultured later in the aeration tank where air is pumped, not in the grit chamber. Skimming of oily scum happens in the settling tank, not the grit chamber, so the purpose stated for slowing flow here is specifically grit settling.
- 4. (B)** Bar screens catch large floating debris; without them, that debris travels on and chokes or jams the downstream equipment, which is the first and most direct problem. Dissolved nitrate levels are unaffected by screening because screens never remove dissolved substances. The aeration bacteria are not killed simply by debris passing through, so they do not die immediately. Letting debris through would make water dirtier, never instantly drinkable, so that option is the opposite of what happens.
- 5. (A)** Air is bubbled into the aeration tank precisely so oxygen-loving (aerobic) bacteria can grow and eat organic waste, and the sludge digester is sealed off from air so anaerobic bacteria break down sludge. So the student's statement matches the chapter exactly. Claiming both need air ignores the sealed, air-free digester. Swapping the two reverses reality, since the aeration tank is the one being supplied air. Saying neither uses bacteria is false because both stages are biological.
- 6. (D)** Biogas is the flammable gas formed when anaerobic bacteria break down sludge inside the closed digester. The aeration tank, being supplied air, hosts aerobic activity and does not make biogas. Manure is the dried, digested sludge, not the oily scum skimmed from the top, so that pairing is wrong. Bar screens only physically catch large solids and produce no gas, so they cannot make biogas.
- 7. (B)** Treatment greatly reduces solids and organic load, but the chapter stresses the released water is cleaner, not guaranteed safe to drink. Claiming every impurity and germ is removed overstates what the process achieves. Equating it with bottled drinking water is the same overstatement. Saying it is dirtier than raw sewage is the opposite of the truth, since treatment clearly improves the water.
- 8. (B)** Sewage first meets bar screens that catch plastics (ii), then the grit chamber where sand settles (iii), then the settling tank where oily scum is skimmed off the top (i), and finally aeration where aerobic bacteria consume leftover organic matter (iv). Starting with scum-skimming puts settling before screening, which is impossible. Starting with aeration reverses the whole plant. Putting grit removal before screening is wrong because large debris must be screened out first to protect the grit chamber.
- 9. (C)** Removing gritty, abrasive sand early protects pumps and keeps it out of the organic sludge that is later digested, which is the real engineering reason. Sand is inert, not the most poisonous part of sewage. Biogas comes from bacterial digestion of organic sludge, not from sand. The settling tank is specifically designed to collect settled solids as sludge, so claiming it cannot hold solids is false.

10. (D) The chapter warns that cooking oil should not be poured down drains because it clogs pipes and hampers the treatment process. Calling oil the favourite food that speeds cleaning contradicts that warning. Oil cannot make water sweet or safe to drink. Manure is produced only by drying digested sludge at the plant, not spontaneously from oil in a sewer, so that option is fanciful.

11. (C) Solids that sink form sludge at the bottom, and the scraper moves this sludge for removal. Skimming the floating scum is the surface device's task, not the bottom scraper's. Bubbling air is done in the aeration tank, not by a scraper. Catching large plastics is the bar screen's job at the plant's entrance, so the bottom scraper deals only with settled sludge.

12. (A) Aerobic bacteria depend on the pumped air for oxygen; cut it off and they cannot work well, so organic matter is left poorly treated. Saying they thrive without air contradicts their need for oxygen. Biogas comes from sealed anaerobic digestion of sludge, not from an aeration tank that has merely lost its air. Dissolved organic matter cannot simply be filtered out by gravity, which is why bacteria are used to consume it in the first place.

13. (A) Wastewater leaves homes through sewers, is carried to the treatment plant for cleaning, and the cleaned water is then discharged into a river or water body. Putting the treatment plant before the sewer reverses the collection step. Sending water to the river before treatment skips cleaning entirely, which the system exists to prevent. Cleaned water is not piped back to home taps, and a biogas plant is not the first stop, so that option is wrong.

14. (C) Sewage carries harmful microbes such as the bacteria that cause cholera and typhoid; these are living and cause disease directly. Phosphate salts are dissolved chemicals, not living organisms. Sand is non-living mineral matter. Cooking oil is non-living organic liquid, so among the choices only the micro-organisms are alive and pathogenic.

15. (A) A septic tank is exactly the low-cost, on-site arrangement for a home without sewer connections. A full treatment plant is large, expensive infrastructure, not a single-home solution. Bar screens and grit chambers are individual stages inside a large plant and do nothing useful on their own under one toilet, so neither serves as an on-site sanitation system.

16. (A) After aeration the bacteria settle as activated sludge, and much of it is sent back into the aeration tank so the bacterial population keeps cleaning incoming water. Pouring it straight into a river untreated would defeat the purpose of treatment. Activated sludge sinks and is not skimmed as floating scum. Bar screens act on incoming large debris at the entrance, not on settled bacterial sludge, so that option is wrong.

17. (B) Treatment depends on living bacteria, and toxic chemicals can poison and kill them, crippling the cleaning and letting harm pass into rivers. Chemicals do not make the purely mechanical bar screens work faster. They cannot instantly produce safe manure, which only comes from properly digested and dried sludge. They certainly do not boost the drinking-water supply, so the real danger is the death of the helpful microbes.

18. (A) Bar screens are coarse physical filters that stop big floating debris but cannot touch substances that are dissolved or microscopic. They cannot remove dissolved nitrates, which slip through easily. They are not disinfectants and kill no microbes. They certainly do not make sewage completely impurity-free, since fine and dissolved matter remain for later stages.

19. (B) Cholera and typhoid spread when sewage contaminates drinking water; safe toilets and sewers keep waste separated from clean water, breaking that path. Adding nutrients to drinking water is not what sanitation does and would not protect health. Sanitation lowers, not raises, microbes reaching taps. Brighter biogas flames have nothing to do with stopping waterborne disease, so the protective link is keeping sewage out of drinking water.

20. (B) Large objects go first at the bar screens (i), then heavy grit settles in its chamber (iii), then sludge and scum are separated in the settling tank (iv), and finally aerobic bacteria handle leftover organic matter (ii). Starting with bacterial treatment reverses the whole plant. Putting grit before screening is wrong because debris must be screened first. Starting with sludge and scum places settling before screening and grit removal, which cannot happen.

21. (C) Digested sludge is dried into manure for fields, and the biogas given off is burned as fuel. Calling the leftover solid scum is wrong because scum is the floating oil skimmed far earlier, and biogas is not a drinking-water disinfectant. Swapping the two so the solid becomes fuel and the gas becomes manure is backwards. Grit is removed at the start and is not a digester product, so that option fails.

22. (C) Organic contaminants come from living matter; food scraps, faeces and cooking oil all qualify. Dissolved iron salts are inorganic minerals, not organic matter, so they are the odd one out. The other three are classic examples of the organic load in domestic sewage, which is why only the iron salts answer the 'not organic' question.

23. (A) The grit chamber takes out only dense sand and pebbles; lighter suspended solids, oily scum and dissolved impurities go on to the settling and aeration stages. Claiming all solids are gone ignores those later-removed materials. Saying only pure water remains is the same overstatement. The grit chamber removes solids rather than adding any, so the 'adds new solids' option is also false.

24. (D) Medicines and chemicals can poison the living bacteria essential to treatment, so they should not be flushed. They do not energise the bacteria; they damage them. They cannot convert sewage into drinking water. Bar screens are mechanical and gain nothing from such chemicals, so harming the helpful bacteria is the genuine reason to keep these items out of the drain.

25. (D) The digester works without oxygen using anaerobic bacteria, while the aeration tank is bubbled with air for aerobic bacteria, so they are opposites in their air supply. Saying the digester gets air and the aeration tank is sealed reverses the facts. Claiming both are sealed ignores the aeration tank's air pump. Claiming both are aerated ignores the sealed digester, so only the stated contrast is right.

26. (D) Human waste left in the open carries disease germs that rain can wash into wells, ponds and rivers people use, spreading illness. Open waste does not become safe biogas, which needs sealed anaerobic digestion. It contaminates rather than increases clean water. There are no helpful aerobic bacteria living in taps that this could feed, so the real hazard is contamination of water sources.

27. (B) After grit removal, the settling tank lets finer suspended solids sink and collect as sludge. Bar screens are coarse and only catch large objects, not fine suspended particles. The digester acts on sludge already collected, it is not a sieve for incoming suspended solids. The aeration tank's air is meant to feed aerobic bacteria, not to settle solids, so the settling tank is the correct stage.

28. (C) Air is pumped into the aeration tank so aerobic bacteria work there, while the digester is sealed so anaerobic bacteria work without air. Reversing the two contradicts the air supply that defines each. Saying both use air ignores the sealed digester. Saying both are air-free ignores the air pump on the aeration tank, so the matched pairing is the only correct one.

29. (D) In a vermi-processing toilet, earthworms break down human excreta into compost-like manure, avoiding pollution. Flushing waste straight into a river is the opposite of environment-friendly. The toilet does not burn waste, and biogas is not made this way. There is no air pump growing aerobic bacteria in such a toilet, so the earthworm-to-manure description is the right one.

30. (A) Earlier steps remove solids and much organic matter but not all germs, so a disinfectant like chlorine is used to kill remaining harmful microbes before discharge. Chlorine is not added to enrich water with nitrates. Settling of sand was already done in the grit chamber, not by chlorination. Biogas comes from anaerobic digestion of sludge, not from chlorinating clarified water, so disinfection of microbes is the purpose.

31. (C) Used water arrives from homes, from industries and institutions, and from farms, so all three feed the sewer, showing sewage has mixed sources. Limiting it to household

toilets ignores industrial and agricultural inputs. Sewage is contaminated used water, not clean rooftop rainwater. The plant cleans sewage, it does not create it, so the multi-source description is correct.

32. (B) The settling tank is recognised by quiet conditions where sludge sinks and scum floats, with no aeration. Vigorous air bubbling to grow bacteria is the signature of the aeration tank, not the settling tank. Catching plastic bags on metal bars describes bar screens. Collecting flammable gas from a sealed dome describes the digester, so the sludge-plus-scum-with-no-air picture identifies the settling tank.

33. (B) Nitrates and phosphates act as nutrients that can over-fertilise a water body and upset its natural balance, which is why sewage must be treated first. They do not make lake water safe to drink. Being dissolved, they do not settle out as sand. They do not spontaneously become biogas in open lake water, which would need sealed anaerobic digestion, so the nutrient-overload effect is the correct consequence.

34. (C) Scum is the floating layer of oils and grease removed by skimming the surface. Heavy material scraped from the bottom is sludge, not scum. The flammable gas from the digester is biogas. Dried sludge spread on fields is manure, so among the choices only the floating, skimmed oily layer correctly defines scum.

35. (D) Untreated sewage pollutes rivers and spreads diseases like cholera and typhoid, so it is cleaned first to protect water and health. Dumping does not usefully deepen a river. Sewage treatment is not run to supply construction sand. It is not about heating river water, so disease prevention and pollution control are the real reasons.

36. (D) The bottom scraper's job is to move settled sludge for removal, so if it jams, sludge piles up at the bottom. Skimming of surface scum is a separate surface device's task, unaffected by the bottom scraper. The aeration tank's air supply is a different system entirely. Bar screens at the entrance are unrelated mechanical bars, so the direct effect is sludge accumulating at the settling tank's floor.

37. (C) Plain rinse water carries no toxic chemicals or pipe-coating grease, so it is the safest to drain. Paint thinner is a harsh chemical that can kill helpful bacteria. Cooking oil coats and clogs pipes. Expired medicines contain chemicals that damage the treatment microbes, so among the four only the plain water is essentially harmless to the system.

38. (B) Sludge goes to the sealed digester for anaerobic breakdown, while the clarified water is sent to the aeration tank to be aerated for aerobic bacteria, two separate streams in two conditions. Treating both together in one air-free tank ignores the aeration step. Swapping so the water is digested without air and the sludge aerated reverses the design. Bacteria act in the digester and aeration tank, not on the mechanical bar screens, so only the stated split is correct.

39. (C) A chemical toilet adds chemicals that decompose waste and control smell and germs, a handy on-site option where there is no sewer connection. It does not produce drinking water. It does not run an air pump to grow aerobic bacteria. It is not designed to collect biogas, so the chemical breakdown of waste for sanitation is its real purpose.

40. (D) Treatment removes solids and consumes organic matter, so the outlet water is much cleaner than the raw inlet sewage. It is not dirtier; that would mean treatment failed. The two are not identical, since treatment clearly changes the water. Yet 'cleaner' does not mean 'guaranteed drinkable', so claiming it is safe to drink overstates the result; the correct point is that it is far less laden than the inlet.

41. (A) Even a good plant is undermined if households flush oils, chemicals and solids that clog pipes or poison the treatment bacteria, so home habits matter. More oil does not help the plant; it harms it. Home habits do not rearrange the plant's fixed sequence of steps. Households are not asked to disinfect water themselves, so the reason is that careless dumping burdens and damages the system.

42. (D) Bar screens are metal bars that physically block large debris, with no biology involved. The aeration tank uses aerobic bacteria, which is biological. The digester uses anaerobic bacteria, also biological. The settling of activated sludge involves bacteria too, so among the choices only the bar screen works by a purely mechanical action.

43. (B) Biogas is a gas produced by anaerobic digestion of sludge, whereas scum is the floating oil and grease skimmed off the settling tank, so they are completely different. They are not both a floating oily layer, since biogas is a gas. They are not both gases, since scum is a floatable matter. Biogas is not bottom sludge and scum is not a gas, so the only correct judgement is that they are distinct.

44. (C) Sealing the digester keeps out air so anaerobic bacteria can decompose the sludge and yield biogas. Exposing it to air would favour aerobic, not anaerobic, action and produce no biogas. The digester is about bacterial breakdown, not sun-drying. Plastics are already removed at the bar screens long before, so screening is not the digester's role; the sealed, air-free condition is the key.

45. (A) A surge of sand and grit is exactly what the grit chamber is built to settle out, protecting later stages. The digester handles organic sludge, not incoming sand. Chlorination kills microbes and does nothing for grit. The aeration tank grows bacteria and is not meant for heavy sand, so the grit chamber is the stage that matters here.

46. (C) Grit is dense inorganic sand taken out early in its own chamber, while sludge is the largely organic solid that settles in the later settling tank and can be digested. They are not the same material in one tank. Grit is not a floating oily layer (that is scum) and sludge is not a gas. Grit is mineral, not organic, and sludge is certainly not drinking water, so only the first description holds.

- 47. (A)** Keeping drains and toilets clean prevents blockages and overflows that would expose people to disease-carrying waste, protecting health. It reduces, not increases, microbes reaching homes. It does not involve pouring chemicals into sinks. It has nothing to do with kitchen biogas flames, so preventing exposure to waste is the genuine benefit.
- 48. (D)** The roles are swapped: the sealed digester is air-free and uses anaerobic bacteria, while the aerated tank supplies air for aerobic bacteria. So the statement is wrong on both counts, not correct. It is not the case that only the digester part is wrong while the aeration part is right, since both are reversed. The two tanks do not share the same aerobic bacteria, because their air conditions differ.
- 49. (D)** Treated water is clean enough to be returned to the ground to recharge aquifers, making good use of it even though it is not certified drinking water. Calling it pure drinking water needing no care overstates its quality. It is cleaner, not dirtier, than raw sewage. Recharging groundwater does not generate biogas, so the sensible reuse of cleaner water is the right reason.
- 50. (C)** Flammable gas from a sealed dome is biogas produced by the anaerobic digester. The aeration tank supplies air to bacteria and does not collect flammable gas. Bar screens merely block solids and release no useful gas. The settling tank separates sludge and scum without collecting flammable gas, so the sealed-dome biogas points to the digester.
- 51. (A)** Big debris is screened first, grit settles next, then the settling tank separates sludge and scum, and finally aerobic bacteria consume remaining organic matter. Starting with bacterial digestion reverses the plant. Putting grit before debris is wrong because debris is screened first to protect the grit chamber. Starting with scum places settling before screening and grit removal, which is impossible.
- 52. (B)** Being natural does not stop solid scraps and tea leaves from accumulating and clogging pipes, which blocks sewage flow. They do not dissolve to improve water; they pile up. They cannot become drinking water inside a pipe. There are no aerobic bacteria living in home taps to consume them, so the real trouble is pipe blockage.
- 53. (B)** The chapter stresses treated water is cleaner but not certified drinking water, so a downstream village should not drink it untested. The plant does not make water dirtier on purpose. Treated water has far less sewage load than the inlet, not more. Rivers do not automatically purify everything that enters, so the caution rests on 'cleaner, not guaranteed safe'.
- 54. (C)** The valuable outputs are biogas, burned as fuel, and the dried digested sludge used as manure. The plastics caught by screens and the grit are wastes to be discarded, not useful products. Scum is not a cooking fuel and grit is not a fertiliser. Raw untreated sludge dumped in a river is pollution, not a benefit, so only the biogas-and-manure pair is genuinely useful.

55. (A) Removing gritty sand early keeps it from wearing out aeration equipment and burying the helpful bacteria. Sand is inert and is not food for aerobic bacteria. The aeration tank is built for biological treatment, not for removing grit, so it would not handle grit better. Grit settles by gravity in its chamber and does not need aerating, so protecting the later aeration stage is the reason.

56. (D) Activated sludge is the settled clump of aerobic bacteria formed in the aeration tank after they consume organic waste, and much is recycled back. It is not the screened plastic debris, which is mechanical waste. It is not the digester's biogas, which is a gas. It is not the inert grit from the grit chamber, so the bacterial settled mass is the right description.

57. (B) Typhoid and cholera spread through water contaminated by sewage, so poor sanitation that lets sewage mix with drinking water is the likely cause. Producing biogas does not cause these diseases. Recharging groundwater with treated water is a safe reuse, not a disease source. Bar screens catching plastics is normal operation and unrelated to outbreaks, so sewage in drinking water is the root cause.

58. (A) Large solids are handled physically by screens and settling, but dissolved organic matter must be broken down biologically by aerobic bacteria in the aeration tank. Bar screens cannot remove dissolved impurities, only large solids. The grit chamber settles sand, not dissolved impurities, and the digester acts on collected sludge. Dissolved organic matter is not skimmed as scum, so only the mechanical-versus-biological split is correct.

59. (D) A septic tank lets solids settle and partly decompose, but the sludge builds up and must be pumped out from time to time for safe disposal or treatment. The settled waste does not become drinking water. A simple home septic tank does not automatically pipe fuel to the kitchen. Solids genuinely accumulate, so the claim that the tank never needs attention is false; periodic emptying is required.

60. (B) Good sanitation keeps disease-causing sewage out of drinking water, preventing illnesses like cholera and typhoid in the first place, which is more effective than treating people after they fall sick. It is not that doctors cannot treat such diseases; prevention simply avoids the suffering. Clean drains are not run to make biogas for hospitals. Blocked drains spread contamination and make water less safe, not more, so prevention through sanitation is the point.